Expressive piano continuation

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Problem and system

We built a local system that extends a solo-piano MIDI idea and exports editable MIDI plus piano audio. The application supports an LSTM, a Transformer trained from scratch and an unmodified pretrained Aria reference. Our measured objective is reproducible continuation with valid note events and a defensible comparison. Audible musical coherence remains a separate listening question.

Data and representation

All 1,276 MAESTRO v3 MIDI files parsed. A conservative derived manifest retained 1,262 performances: 957 training, 134 validation and 171 test. It excluded 13 metadata-ambiguity cases and one integrity case while retaining official split assignments. Known retained composition groups and exact file hashes do not cross splits. This is a metadata/hash audit, not a complete musical-fingerprint leakage proof.

The pinned Aria tokenizer keeps 17,727 vocabulary entries. Piano events encode pitch/velocity, onset and duration using 10 ms quantization and five-second segments. Sustain becomes note duration. Durations above five seconds clip, and export can shorten repeated-pitch endings. Crops preserve complete note groups, stay within a piece and exclude supplied context and padding from loss.

Models and matched training

The LSTM has one 256-unit layer and 9,620,287 parameters. The Transformer has eight layers, width 512, eight heads, SwiGLU width 1,536, pre-LayerNorm, rotary positions and tied embeddings: 36,356,608 parameters. Both full-corpus baselines use context 2,048, BF16, microbatch four, accumulation four, AdamW and 100M valid targets on an RTX 5070 with 12 GB. Checkpoints retain optimizer, crop/RNG and schedule state.

Scratch checkpoint	Scored targets	Validation loss
LSTM baseline	100,003,399	4.316104
Transformer baseline	100,003,399	3.275785
Transformer continued	150,007,166	3.193548
Transformer musical final	160,021,924	3.178622

Loss is mean next-token negative log likelihood in nats on identical 536 fixed validation crops and 1,093,516 targets. The original two baselines have matching budgets but different parameter counts. Later rows add compute. The musical run also changes loss masking, so cumulative targets are lineage accounting rather than equal-objective exposure. No test data informed fitting or selection.

Targeted Transformer improvement

Ordinary continuation inherited weights, Adam and RNGs from the 100M checkpoint and added 50,003,767 targets. Conditional next-pitch accuracy increased from 21.13% to 25.31%. A further run added 10,014,758 scored targets. Half its crops used one global key shift and time scale, preserving chord/interval relationships. Three quarters supplied a two-to-eight-second musical prompt and scored the continuation, with complete-onset boundaries and a half-note cap. The remaining crops kept ordinary next-token loss.

The final musical checkpoint reduced raw validation loss from 3.193548 to 3.178622, a 0.47% gain. Fixed continuation loss fell from 3.192593 to 3.177317 on 985,823 targets. Conditional pitch accuracy rose from 25.31% to 26.00% on 361,722 real notes. Onset and duration losses improved slightly. These teacher-forced metrics do not measure independently generated long-range coherence. The combined-method screen has no equal-extra-compute control or method ablation.

Generation and evaluation

The local interface accepts a piano MIDI upload or authored demo, preserves the prompt and generates a constrained event continuation. It reports stopping reason, actual generated-note extent and renderer release tail. Requested seconds bound new note onsets. EOS or remaining-context exhaustion can stop early. FluidSynth and a fixed GeneralUser GS piano render every compared model at matched settings.

All 63 meaningful tests passed, including data/window integrity, causality, accumulation, exact diagnostic recovery, augmentation/mask replay, generation, audio and HTTP behavior. Saved checkpoints replayed their validation losses. The original 18-sample comparison and both 16-sample Transformer suites retain all declared attempts. Checked exports preserve prompts/counts and merge zero note groups. One listener supplied eight earlier A/B judgments: five continued preferences, two original, one unclear. One continued vote followed identity reveal. The new candidate remains unassessed by humans.

Limitations and contributions

Scope is expressive solo piano. Context, quantization and duration limits affect outputs. Pretrained exposure differs and overlap is uncertain. No project Aria LoRA adaptation, final test evaluation, broad listening study or completed human rehearsal is claimed. Browser visual/playback inspection of the latest candidate remained unverified, while file and HTTP checks passed. The app selects the numerically verified musical checkpoint.

Equal contribution allocation, as requested by the team: Mohammed Zahoor Mashahir owns data/tokenization, scratch training and checkpoint/Transformer experiments. Mohammed Ibrahim owns the Aria baseline, rendering, application, listening pages and submission packaging. Both share results interpretation, documentation and demonstration preparation. Development used an AI coding assistant; the repository preserves actual experiment evidence.

Reproducibility and sources

Repository: github.com/zahoormzm/expressive-piano-continuation (private). README documents Windows setup, source/data hashes and recovery. A private release supplies checkpoints and evidence. The demo launches with the configured environment and scripts/serve_app.py. Faculty/TA access and rehearsal require completion by the team.

Sources: Hawthorne et al., MAESTRO, ICLR 2019 (magenta.tensorflow.org/datasets/maestro, CC BY-NC-SA 4.0). Huang et al., Music Transformer, ICLR 2019 (arxiv.org/abs/1809.04281). Bradshaw et al., Aria, ISMIR 2025 (arxiv.org/abs/2506.23869). Local measured evidence: docs/full-training-results.json and docs/transformer-musical-training.json. Aria and renderer/asset licenses are attributed in README.